

1 Data Generation

1.1 Pattern Matching

A *pattern* is a set.

1.1.1 Example

Let $a \circ b = [a_1, \dots, a_n, b_1, \dots, b_m]$, where $a = [a_1, \dots, a_n]$ and $b = [b_1, \dots, b_m]$.

$$\begin{aligned}\mathfrak{E} &:= \mathfrak{T} \cup \{\mathfrak{e}_1 \circ \mathfrak{o} \circ \mathfrak{e}_2 \mid \mathfrak{o} \in \{["+"], ["-"]\} \wedge \mathfrak{e}_1, \mathfrak{e}_2 \in \mathfrak{E}\} \\ \mathfrak{T} &:= \mathfrak{F} \cup \{\mathfrak{t}_1 \circ \mathfrak{o} \circ \mathfrak{t}_2 \mid \mathfrak{o} \in \{["\times"], ["\div"]\} \wedge \mathfrak{t}_1, \mathfrak{t}_2 \in \mathfrak{T}\} \\ \mathfrak{F} &:= \mathfrak{N} \cup \{["("] \circ \mathfrak{e} \circ [")"] \mid \mathfrak{e} \in \mathfrak{E}\} \\ \mathfrak{N} &:= N \cup N^* \\ N &:= \{ "0", \dots, "9" \}\end{aligned}$$

What is the difference between pattern matching and tokenwise parsing?

2 gen.jl and pattern.jl

A *grimoire* Φ of a semigroup G , a set of *spells* $\mathfrak{S}_\Phi \subseteq G$ and a set of *movements* $\mathfrak{A} \subseteq L_\Phi$ is a map in $L_\Phi \rightarrow \mathcal{P}(\mathfrak{S}_\Phi^{<\omega})$ such that $\mathfrak{s}_1 * \dots * \mathfrak{s}_n = \mathfrak{s}$ – where $*$ is the binary operator belonging to G – for each $[\mathfrak{s}_1, \dots, \mathfrak{s}_n] \in \Phi(\mathfrak{s})$ and $l \in \mathfrak{S}_\Phi \setminus \mathfrak{A}$.

Let $G := (\langle \mathcal{G} \rangle, *)$ be a semigroup. Let $\Delta : \boxed{\Phi} \times G^2 \rightarrow \boxed{\Phi}$ – where $\boxed{\Phi}$ is the set of all grimoires of a pattern $\mathfrak{P} \subseteq G^{<\omega}$ – be defined for some grimoire Φ by $\forall \mathfrak{s} \in \mathfrak{S}_\Phi \setminus \{\mathfrak{s}_1 * \mathfrak{s}_2\}$ ($\Phi'(l) = \Phi(l)$) and $\Phi'(\mathfrak{s}_1 * \mathfrak{s}_2) = \Phi(\mathfrak{s}_1 * \mathfrak{s}_2) \cup \{[\mathfrak{s}_1, \mathfrak{s}_2]\}$ if $\mathfrak{s}_1 * \mathfrak{s}_2 \in \mathfrak{S}_\Phi$ and $\Phi'(\mathfrak{s}_1 * \mathfrak{s}_2) = \{[\mathfrak{s}_1, \mathfrak{s}_2]\}$ if not – where $\Phi' = \Delta(\Phi, [\mathfrak{s}_1, \mathfrak{s}_2])$ and $\mathfrak{s}_1, \mathfrak{s}_2 \in G$. Note that since $\mathfrak{S}_\Phi = \text{dom}(\Phi)$, then $\mathfrak{S}_{\Phi'} = \mathfrak{S}_\Phi \cup \{\mathfrak{s}_1 * \mathfrak{s}_2\}$ if $\Phi' = \Delta(\Phi, [\mathfrak{s}_1, \mathfrak{s}_2])$.

???? $\Phi : \mathcal{G} \rightarrow \{\emptyset\}$????Now, consider the exploration function $\mathcal{E} :$

$\boxed{\Phi} \times \boxed{\mathcal{C}}^{<\omega} \rightarrow \boxed{\Phi}$ defined by $\mathcal{E}(\Phi, [\mathcal{C}_1, \dots, \mathcal{C}_n]) := \bigcirc_{i=1}^n \Delta(\Phi, \mathcal{C}_i(L_\Phi))$!!!!!!This

is another subgroup!!!, where $\boxed{\mathcal{C}} \subset \mathcal{P}(G) \rightarrow G^2$ and $\mathcal{C}(S) \in S^2$ for each $\mathcal{C} \in \boxed{\mathcal{C}}$ and $S \in \mathcal{P}(G)$. Then !!!!!THIS SHOULD BE A LEMMA!!!! Φ_n is a partial or full representation of the group G , where each element in $L(\Phi_n)$ maps to the set of ordered pairs of elements in G to which it is mapped by $*$.

!!!!!!! therefore, for each output generated by $\mathcal{E}$, one can define a function λ such that: !!!!!!! Let the functions $\lambda, \lambda_{\text{step}} : X_G \rightarrow \mathcal{G}^{<\omega}$ – where

$$X_G = \left\{ [\Phi, \sigma, \Pi, T] \in (L_\Phi \rightarrow \mathcal{P}(G^{<\omega})) \times G^{<\omega} \times \boxed{\Pi} \times \mathcal{P}(G) \mid \sigma \in L_\Phi^{<\omega} \wedge T \subseteq L_\Phi \wedge \forall$$

– be defined by

$$\lambda(\Phi, \sigma, \Pi, T) := \begin{cases} \sigma & : \lambda_{\text{step}}(\Phi, \sigma, \Pi, T) = \sigma \\ \lambda(\Phi, \lambda_{\text{step}}(\Phi, \sigma, \Pi, T), \Pi, T) & : \text{otherwise} \end{cases}$$

and

$$\lambda_{\text{step}}(\Phi, \sigma, \Pi, T) := \bigotimes_{i=1}^{|\sigma|} \left(\begin{cases} \sigma_i & : \sigma_i \in T \\ \Pi(\Phi(\sigma_i)) & : \text{otherwise} \end{cases} \right),$$

where $\boxed{\Pi} \subset \mathcal{P}(G^{<\omega}) \rightarrow G^{<\omega}$ is a set of maps wherein $\Pi(S) \in S$ for each $\Pi \in \boxed{\Pi}$ and $S \in \mathcal{P}(G^{<\omega})$ hold.

!!!!!!BEVISA FAKTUM OM DEN GENERERADE STRUKTUREN.

SKRIV EXEMPEL FÖR ATT DEMONSTRERA HUR EN GRIMOIRE BYGGS UPP AV $\mathcal{E}$, OCH SEDAN HUR λ GENERERAR EN LÖSNING – T.EX. FÖR EN PERMUTATIONSGRUPP (ELLER NÅGON ANNAN SYNTAX).!!!!!!

BEVISA ATT GRIMOIRE-UTFORSKNING ÄR EN SEMIGRUPP (OCH ATT EN GRIMOIRE DÄRMED KAN DEFINIERAS MED DEN)